

Hands on 3 - Branching & Merging

Step 1: Create a New Repository

Commands:

```
mkdir GitBranchDemo
cd GitBranchDemo
git init
ls -la
```

Explanation:

Creates a new Git repository and verifies the .git folder.

Step 2: Create the First File

Commands:

```
echo "Welcome to Git Branching" > welcome.txt
ls
cat welcome.txt
git status
```

Explanation:

Creates welcome.txt and checks that Git detects it.

Step 3: Stage the File

Commands:

```
git add welcome.txt
git status
```

Explanation:

Moves welcome.txt to the staging area.

Step 4: First Commit

Command:

```
git commit -m "Initial commit"
```

Explanation:

Creates the first commit.

Step 5: Create Branch

Commands:

```
git branch GitNewBranch
git branch
```

Explanation:

Creates a new branch and lists all branches.

Step 6: Switch Branch

Commands:

```
git checkout GitNewBranch  
git branch
```

Explanation:

Switches to GitNewBranch.

Step 7: Add File in Branch

Commands:

```
echo "This file is created in GitNewBranch" > branch.txt  
ls  
cat branch.txt  
git status
```

Explanation:

Creates a file only in GitNewBranch.

Step 8: Commit Branch Changes

Commands:

```
git add branch.txt  
git commit -m "Added branch.txt in GitNewBranch"
```

Explanation:

Commits the new file to GitNewBranch.

Step 9: Switch to Master

Commands:

```
git checkout master  
ls
```

Explanation:

branch.txt disappears because it exists only in GitNewBranch.

Step 10: Compare Branches

Command:

```
git diff master GitNewBranch
```

Explanation:

Shows differences between branches.

Step 11: Merge

Commands:

```
git merge GitNewBranch  
ls
```

Explanation:

Merges GitNewBranch into master.

Step 12: View History

Command:

```
git log --oneline --graph --decorate
```

Explanation:

Displays commit history.

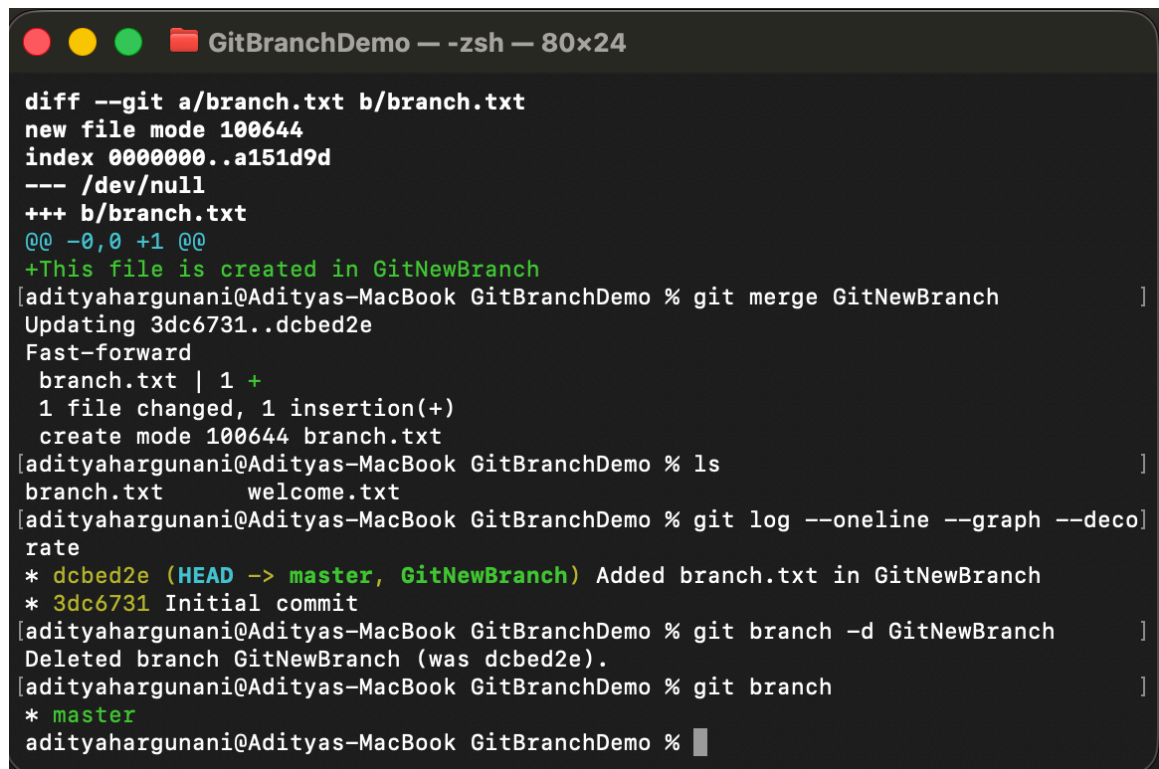
Step 13: Delete Branch

Commands:

```
git branch -d GitNewBranch  
git branch
```

Explanation:

Deletes GitNewBranch after merging.



```
GitBranchDemo — -zsh — 80x24  
  
diff --git a/branch.txt b/branch.txt  
new file mode 100644  
index 0000000..a151d9d  
--- /dev/null  
+++ b/branch.txt  
@@ -0,0 +1 @@  
+This file is created in GitNewBranch  
[adityahargunani@Adityas-MacBook GitBranchDemo % git merge GitNewBranch  
Updating 3dc6731..dcbed2e  
Fast-forward  
  branch.txt | 1 +  
  1 file changed, 1 insertion(+)  
  create mode 100644 branch.txt  
[adityahargunani@Adityas-MacBook GitBranchDemo % ls  
branch.txt      welcome.txt  
[adityahargunani@Adityas-MacBook GitBranchDemo % git log --oneline --graph --deco  
rate  
* dcbed2e (HEAD -> master, GitNewBranch) Added branch.txt in GitNewBranch  
* 3dc6731 Initial commit  
[adityahargunani@Adityas-MacBook GitBranchDemo % git branch -d GitNewBranch  
Deleted branch GitNewBranch (was dcbed2e).  
[adityahargunani@Adityas-MacBook GitBranchDemo % git branch  
* master  
adityahargunani@Adityas-MacBook GitBranchDemo %
```